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Electro-optic modulator with ultra-low residual amplitude modulation for frequency modulation and laser stabilization

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The reduction of the residual amplitude modulation (RAM) induced by electro-optic modulation is essential for many applications of frequency modulation spectroscopy requiring a lower system noise floor. Here, we demonstrate a simple passive approach employing an electro-optic modulator (EOM) cut at Brewster's angle. The proposed EOM exhibits a RAM of a few parts per million, which is comparable with that achieved by a common EOM under critical active temperature and bias voltage controls. The frequency instability of a 10 cm cavity-stabilized laser induced by the RAM effect of the proposed EOM is below 3×10^{-17} for integration times from 1 to 1000 s, and below 4×10^{-16} for comprehensive noise contributions for integration times from 1 to 100 s. © 2016 Optical Society of America

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Frequency modulation (FM) spectroscopy [1] is a technique widely used in many highly precise detection systems, such as those employed in gravitational-wave detection [2–4], trace gas sensing in medicine and atmospheric sciences [5–7], laser precision spectroscopy [1,8], and ultra-stable lasers [9–16]. Electro-optic modulators (EOMs) are components employed for modulating the phase/frequency of lasers in the FM technique. However, the phase/frequency modulation provided by an EOM is always accompanied with residual amplitude modulation (RAM). RAM results in two modulation sidebands that are not exactly equal in magnitude and/or opposite in phase. This asymmetry produces an offset on the demodulated signal of the FM technique, which degrades the accuracy of detection systems employing FM spectroscopy and limits its application to detection systems.

RAM is introduced in various ways [17–19], among which the parasitic etalon effect [17] and the polarization rotation

effect [18] usually dominate the RAM level at the modulation frequency. The parasitic etalon effect is due to reflection from the parallel surfaces of the EOM owing to imperfect antireflection coatings [17], and the polarization rotation effect is caused by the birefringence of the electro-optic (EO) crystal [18]. Both effects are sensitive to environmental noises, such as temperature fluctuations and mechanical vibrations, leading to an unstable offset on the demodulated signal of the FM technique.

Fortunately, this offset goes to zero under certain symmetrical operation conditions, and, thus, a straightforward solution is to actively control the system to operate around these symmetry points [18,20–23]. This method has been employed for more than 30 years. More specifically, Wong and Hall analyzed the RAM caused by the polarization rotation of the EOM, and found that applying a bias voltage could compensate the natural birefringence of the EO crystal, and, thus, suppress the polarization rotation effect [18]. Meanwhile, Gehrtz *et al.* employed a dual-beam method to reject the RAM in a postoptical detector [24]. A few research groups have subsequently adopted similar approaches to reduce RAM below the 10^{-4} level [20–24] by actively controlling the electric field acting on the EOM and the temperature of the EOM [20–22], or by applying other active devices such as an acoustic optical modulator in the optical path [23]. However, these active approaches are difficult to realize, because several parameters must be stabilized simultaneously and the references of these control loops must be sufficiently accurate. Moreover, a mechanism for accurately distinguishing noise sources is required to maintain a low RAM level over the long term because of the cross talk between control loops [22].

Alternatively, a few passive solutions for reducing RAM have been developed. Setting a polarizer with a high extinction ratio in front of the EOM can reduce the polarization rotation effect [25]; however, the added polarizer is also sensitive to environmental conditions. It has been shown that tilting a cuboid EOM off the vertical incidence can reduce the parasitic etalon effect [24]; however, tilting the EOM leads to additional polarization rotation, while the tilt angle is limited to the aperture of the EOM. Recently, a 2.85° wedge-faced trapezoidal crystal

with good antireflection (AR) coatings was used to construct the EOM for the detection system of the Enhanced Laser Interferometer Gravitational-Wave Observatory [26]. The modification was shown to significantly suppress both the parasitic etalon effect and the polarization rotation effect [27]. Employing an equivalent technique, a 4° wedge-end trapezoidal crystal was used to generate a squeezed state, resulting in a reduction of quantum noise fluctuations by a factor of 50 [28]. For an EOM employing a trapezoidal crystal, laser beams with perpendicular polarizations would have a separation in angle; however, the beam passing through the EOM would deflect from its original direction, and, thus, a small change in the angle or in the refractive index can lead to a profound change in the position after traversing a long path. Therefore, the system is sensitive to vibrations and temperature changes. While an EOM using a rhomboid-shaped crystal would be immune to these influences because of the null deflection angle, a rhomboid-shaped crystal is not as effective as a trapezoidal crystal for the separation of perpendicularly polarized beams, which is a key consideration for suppressing the polarization rotation effect. Therefore, we employ a trade-off solution in this Letter by utilizing a large tilt-angle rhomboid-shaped EOM to separate beams sufficiently, while retaining good insensitivity to environmental variations, and obtain a RAM level of several parts per million (ppm).

The EOM is shown in Fig. 1, where an EO crystal composed of LiNbO_3 is employed due to its large electro-optic coefficient. The optic axis of the crystal is defined as the z axis, which is also the direction of the applied electric field. Light propagates along the x axis. The EO crystal is cut from a $50 \text{ mm} \times 5 \text{ mm} \times 5 \text{ mm}$ ($x \times y \times z$) cuboid LiNbO_3 crystal, where the cross-sectional size is sufficiently large for passing the $250 \text{ }\mu\text{m}$ diameter laser beam employed in our experiment. The acute angle θ is the Brewster's angle of the material for a given laser wavelength λ ; therefore, the incidence angle and the emergence angle are Brewster angles. In the present work, $\lambda = 1555 \text{ nm}$, which yields a Brewster's angle of 65.66° .

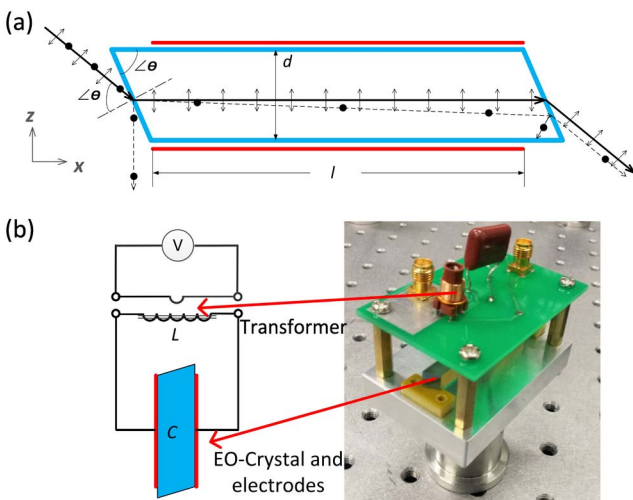


Fig. 1. Custom-made EOM with edges cut along Brewster's angle. (a) Top view of the EO crystal, where θ is Brewster's angle. The dots and double-headed arrows represent the electric fields of the ordinary and extraordinary laser beam components, respectively, and red lines represent the electrodes. (b) Schematic of the impedance-matching tank circuits and image of the EOM.

We choose the extraordinary component of the laser beam (i.e., polarized in the z direction) for the following utilization. This component passes along the x axis of the EO crystal, parallel to the electrodes. The ordinary component of the beam (i.e., polarized in the y direction) is separated by 0.74° from the extraordinary component owing to the natural birefringence. At the output of the EO crystal, the two parallel beams are separated by a distance of 0.28 mm , which is sufficient to guarantee that the ordinary beam component keeps off resonance.

By employing Brewster angles, the incident and emerging extraordinary light component undergoes no reflection, so that no AR coating is required. This not only saves time and expense, but also avoids problems associated with AR coating damage due to laser burning [29]. The resulting high laser-induced damage threshold of the EOM is advantageous for many applications, e.g., gravitational-wave detection [2] and high-power, high-brightness laser modulation [30]. Although the Brewster's angle employed is designed for a specific laser wavelength, the AR capability of the angled surface is less sensitive to the wavelength than common AR coatings. The calculated insertion loss of the EO crystal is below 0.03% for wavelengths between 1000 and 2000 nm .

The half-wave voltage, i.e., the applied voltage producing a phase shift of π on the light passing through the EOM, is proportional to the length-width ratio, as follows:

$$V_\pi = \frac{\lambda d}{n_e^3 \gamma_e l}. \quad (1)$$

Here, n_e is the refractive index of the extraordinary component, γ_e is the electro-optic coefficient, d (5 mm) is the distance between the electrodes, and l (45.5 mm) is the length of the electrodes. The value of V_π is calculated to be 540 V , which is difficult to apply directly; therefore, we constructed a tank circuit, including a transformer (12:1) and a serial inductor-capacitor resonant circuit. The capacitance of the EO crystal was calculated to be about 10 pF , and a magnetic core type inductor was employed. The return loss of the EOM is -32 dB at a resonant frequency of 35.9 MHz , and the half-wave voltage is thereby reduced to 25 V at the input terminal for a standard $50 \text{ }\Omega$ impedance-matching condition.

Figure 2 shows the Pound-Drever-Hall (PDH) frequency stabilization scheme [16] employed in the experiments. To evaluate the performance of the proposed EOM, we compared

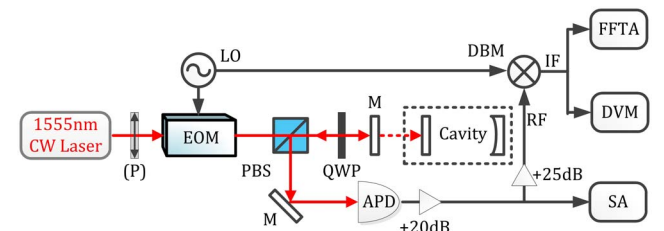


Fig. 2. Experimental scheme, where the red lines represent optical pathways and the black lines are electrical signal pathways. The polarizer (P) is employed only in front of the commercial EOM. The individual components are as follows: CW, continuous wave; LO, local oscillator; PBS, polarization beam splitter; QWP, quarter-wave plate; M, mirror; APD, avalanche photodetector; DBM, double-balanced mixer; FFTA, fast Fourier transform analyzer; DVM, digital voltmeter; SA, spectrum analyzer.

the device under test (DUT) with a commercial EOM (New Focus-4003). A high extinction ratio ($>10^4$) polarizer is inserted in front of the commercial EOM to reduce the polarization rotation effect. A 0.2 mW laser (NKT Photonics Koheras Adjustik-E15) is passed through the EOM, which modulates the light with a depth of 0.7 rad. Subsequently, the laser beam is reflected by a mirror or the cavity at an off-resonance condition, and then focused into an avalanche photodiode (APD). The photocurrent of the APD is about 0.5 mA. The signal at the modulation frequency from the APD is amplified by two cascaded radio-frequency (RF) amplifiers with gains of 20 and 25 dB. A spectrum analyzer (Anritsu MS2692A) is employed for measuring the magnitude of the RAM. We obtain the demodulated signal, i.e., the error signal of the PDH locking system, using a double-balanced mixer. The conversion loss of the mixer from the RF port to the intermediate-frequency (IF) port is about 6 dB. The phase between the two inputs of the mixer has been adjusted to yield a maximum output for the error signal. The IF port of the mixer is connected to a fast Fourier transform analyzer (Stanford SR785) and a digital voltmeter (DVM; Keithley 2000).

To measure the pure RAM of the EOM, we detect the mirror reflected laser. Figure 3(a) shows the RAM signal at the modulation frequencies of the DUT and the commercial EOM. Here, the normalized RAM level is defined as the

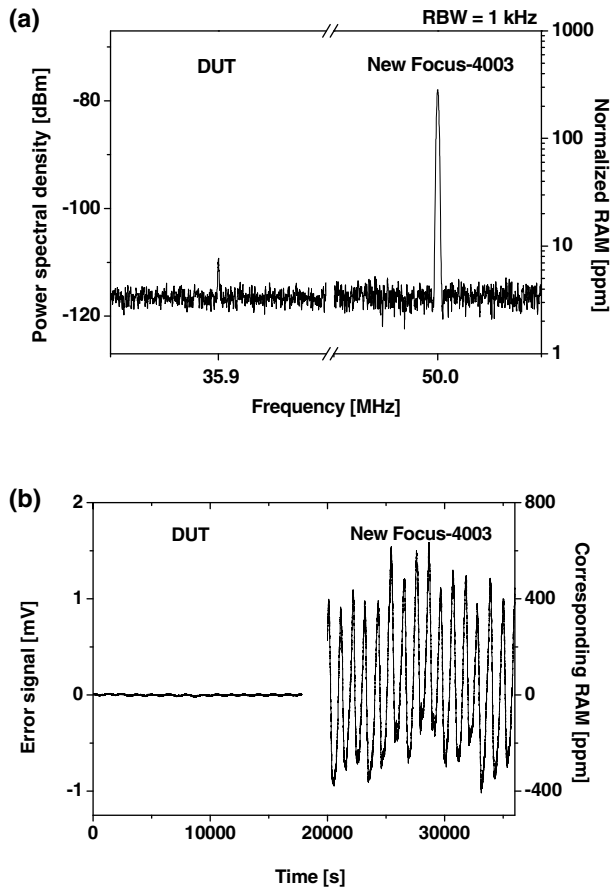


Fig. 3. (a) RAM signal in the frequency domain, and (b) error signal of an open-loop PDH frequency-stabilization system with a New Focus-4003 EOM and the DUT.

amplitude-modulation depth/phase-modulation depth [24]. The RAM level of the DUT is about 7 ppm, which is 40 times lower than that of the commercial EOM. For the FM technique, the long-term fluctuation is in most cases more important than the magnitude of the RAM, because such fluctuations cause an unstable demodulated signal that degrades the stability of the entire system. Figure 3(b) shows the measured RAM-induced error signals of these two EOMs recorded by the DVM over a period of more than 4 h. The observed error signal fluctuations with a periodicity of about 20 min are attributed to temperature fluctuations. Clearly, the DUT is considerably more insensitive to temperature than the commercial EOM.

For cavity-based ultra-stable lasers [9–16] using the PDH locking technique, an unstable error signal contributes to the frequency instability of the laser according to the following relation:

$$\sigma_y = \frac{\sigma_e}{D} \cdot \frac{1}{\nu}. \quad (2)$$

Here, σ_e is the Allan deviation of the error signal [see Fig. 3(b)] obtained by DVM, D is the PDH discrimination of 1×10^{-3} V/Hz, and ν is the frequency of the optical carrier (~ 193 THz).

Figure 4 shows noise effects on frequency instabilities. In the mirror reflection condition, denoted by the filled triangles and circles for the DUT and commercial EOM, respectively, the values of σ_e represent only the RAM-induced frequency instabilities. That of the DUT is about 1×10^{-17} at a 1 s integration time, which is about 6 times lower than that of the commercial EOM at a 1 s integration time, 100 times lower at a 10 s integration time, and 200 times lower at a 100 s integration time. Moreover, the DUT results are 5–30 times lower for integration times of 1–1000 s in comparison with a previously reported active method [22]. To more realistically simulate

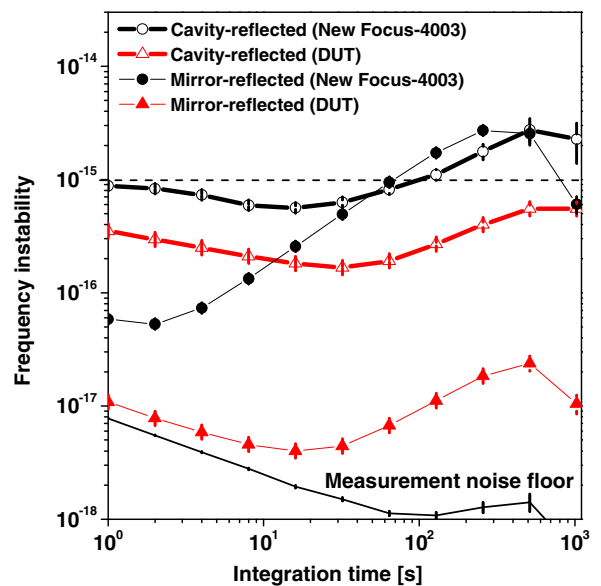


Fig. 4. Noise effects of the 10 cm cavity-stabilized ultra-stable laser. The horizontal dashed line is the thermal noise effect of a 10 cm cavity with ULE mirrors.

the technical noise of an ultra-stable laser system, we generated an error signal reflected from the cavity instead of the mirror, and adjusted the laser frequency away from the resonate point. In this condition, error signal fluctuations are attributed to comprehensive noises, which include not only the pure RAM due to EOM modulation but also some other effects. We assume that the main contribution derives from the free-running laser interacting with the high-order modes of the cavity [31]. These data points are given in Fig. 4 by the open triangles and circles for the system employing the DUT and commercial EOM, respectively. The frequency instability is about 3.5×10^{-16} at an integration time of 1 s for the system employing the DUT. The thermodynamic effect of the 10 cm ultra-stable cavity, which consists of an ultra-low expansion glass (ULE) spacer and ULE-based mirrors, is calculated to be about 1×10^{-15} [32]. Therefore, the comprehensive noise effect can be ignored over the full measurement range. The instability level using the commercial EOM is 3–6 times higher, and further suppression may be required for obtaining stability over long periods because the instability is greater than the thermodynamic effect. This is why active controls are applied on EOM working conditions, particularly for the temperature [20–22]. Temperature fluctuations can change laser polarization, RAM level, and optical coupling and all of these eventually lead to fluctuations in the error signal. Obvious risings in the frequency instability data shown in Fig. 4 over a period of a few hundred seconds are due to temperature fluctuations of the laboratory (2°C peak-to-peak temperature fluctuation). Note that this experiment was conducted in an open laboratory space, and performance can be further improved by applying environmental noise isolation.

In the PDH frequency-stabilized laser system, the measured frequency instability obtained by detecting the cavity-reflected laser provides an upper bound estimation of the overall noise effects, excluding noise owing to fluctuations in the cavity length. It is worth noting that state-of-the-art ultra-stable lasers attain a frequency instability as low as 8×10^{-17} at an integration time of 1 s with a 48 cm cavity [14] and a value of about 1×10^{-16} with a 21 cm cryogenic cavity [12,15]. The performance of ultra-stable lasers is being continuously improved, and both active and passive approaches may need to be combined to further suppress noise effects.

In conclusion, we demonstrated an EOM with an ultra-low RAM level fabricated using surfaces cut at Brewster's angle. The RAM level was suppressed to a few ppm and was shown to be insensitive to environmental noises. In a 10 cm cavity-stabilized ultra-stable laser, the frequency instability induced by the RAM effect of the EOM was less than 3×10^{-17} for integration times from 1 to 1000 s, and below 4×10^{-16} for comprehensive noise contributions with integration times from 1 to 100 s. In comparison with the results obtained using a commercial EOM, these noise effects are a few times to one hundred times lower for integration times from 1 to 1000 s. The comprehensive noise effect was well below the thermodynamic effect of the cavity without any additional control and noise isolation. This result is comparable to that obtained using complex active control. The proposed EOM is a simple solution for attaining a low detection noise floor that would be applicable to other systems employing the FM technique.

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